

Lecture Notes on Agentic AI

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2 Agent Taxonomies: A Framework for Understanding Intelligent Systems

This lecture has the following learning objectives:

1. Formalize the agent-environment interaction loop introduced in Lecture 1
2. Classify agents into five classical categories: simple reflex, model-based reflex, goal-based, utility-based, and learning agents
3. Identify concrete examples of each category
4. Connect the *learning agent* category to reinforcement learning, which you studied last semester
5. Position modern agentic AI (LLM-based agents) within this classical taxonomy

2.1 The Agent-Environment Loop

Recall from Lecture 1 that an agent perceives its environment through sensors and acts upon it through actuators. Figure 1 makes this loop explicit. This loop is the one piece of structure that *every* agent in this lecture shares – what differs between agent types is entirely *what happens inside the agent box*.

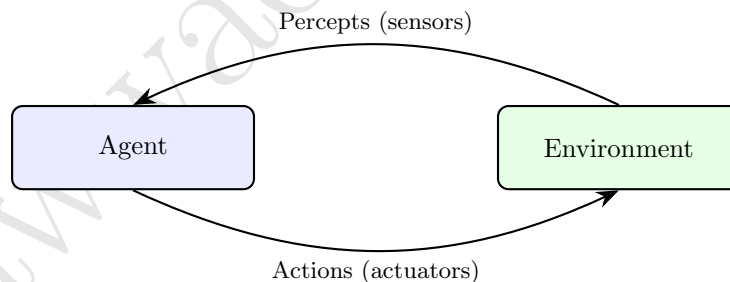


Figure 1: The generic agent-environment interaction loop.

The five agent types below are distinguished by *how much of the past* they retain, and *how far ahead* they reason, before choosing an action.

2.2 Simple Reflex Agents

Definition

A **simple reflex agent** selects an action based only on the *current* percept, using a fixed set of condition-action rules. It has no memory of the past and does not consider the consequences of its action beyond the present moment.

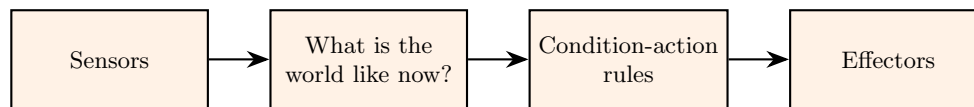


Figure 2: Internal architecture of a simple reflex agent: action depends only on the current percept.

Example

Thermostat. A thermostat perceives the current temperature and applies a single rule: *if temperature < target, turn on heating; otherwise, turn it off*. It does not remember yesterday's temperature and does not plan ahead – it simply reacts to what it senses right now.

2.3 Model-Based Reflex Agents

A simple reflex agent breaks down as soon as the correct action depends on something the agent cannot currently perceive. Model-based reflex agents fix this by maintaining **internal state** – a model of how the world evolves and how the agent's own actions affect it.

Definition

A **model-based reflex agent** maintains internal state that tracks aspects of the world it cannot currently observe, updating this state based on how it believes the world evolves and how its own past actions affected it.

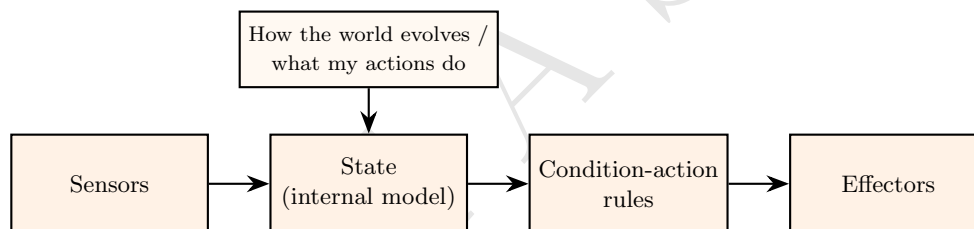


Figure 3: Internal architecture of a model-based reflex agent: an internal state is maintained and updated over time.

Example

Robot vacuum. A robot vacuum cannot see the whole house at once, but it keeps an internal map of which rooms it has already cleaned. When deciding where to go next, it consults this internal state, not just what its sensors currently detect.

2.4 Goal-Based Agents

Reflex agents, even model-based ones, only ask “what rule applies right now?” They do not ask “what do I ultimately want to achieve, and which action moves me closer to it?” Goal-based agents make this explicit.

Definition

A **goal-based agent** selects actions by considering future consequences and choosing the action (or sequence of actions) that is expected to achieve an explicit goal.

Example

GPS navigation. When you enter a destination, the app does not simply react to your current location – it searches over possible routes and selects a sequence of turns that leads to your goal (the destination). This requires *planning*, not just reflex.

2.5 Utility-Based Agents

A goal-based agent can tell you *whether* a route reaches the destination, but not whether it is the *best* route if there are several that all work. Utility-based agents add a way to compare between multiple ways of achieving a goal.

Definition

A **utility-based agent** selects actions by maximizing a **utility function** – a numeric measure of how desirable a given outcome is – rather than merely checking whether a goal is satisfied.

Example

Self-driving car. Reaching the destination is the goal, but the car must also balance speed, fuel consumption, passenger comfort, and safety. A utility function combines these competing factors into a single score, and the car chooses the action that maximizes it – not simply the first action that eventually gets there.

2.6 Learning Agents

Every agent type so far assumes the designer already knows the right rules, model, goals, or utility function in advance. Learning agents remove this assumption: they **improve their own behavior from experience**.

Definition

A **learning agent** improves its performance over time by interacting with its environment, receiving feedback (often called a *reward*), and adjusting its future behavior accordingly.

Question

Does this sound familiar? It should – this is exactly the setting of **reinforcement learning**, which you studied last semester. An RL agent (like one trained to play a game, or a robotic control policy) is a learning agent: it does not start with a hand-coded policy or utility function – it discovers one through trial, error, and reward.

Example

AlphaGo. AlphaGo was not given hand-coded rules for what a “good” Go move looks like. It learned, through millions of games of experience (including playing against itself), which moves tend to lead to winning outcomes.

2.7 Comparing the Five Agent Types

Table 1 summarizes the progression: each agent type adds a capability that the previous one lacked.

2.8 Where Does Agentic AI Fit?

We now return to the question this lecture has been building toward: where does an LLM-based agent – the subject of this entire course – fit into this half-century-old taxonomy?

Agent Type	State?	Plans Ahead?	Learns?	Example
Simple reflex	No	No	No	Thermostat
Model-based reflex	Yes	No	No	Robot vacuum
Goal-based	Yes	Yes	No	GPS navigation
Utility-based	Yes	Yes (optimized)	No	Self-driving car
Learning	Yes	Depends	Yes	AlphaGo, RL policy

Table 1: The classical agent taxonomy.

Example

Agentic AI is not a sixth category – it is a new *implementation* of the existing ones. A modern LLM-based agent typically behaves like a **goal-based** agent (it is given an objective and plans steps toward it), often incorporates **utility-like** reasoning (choosing between multiple valid plans), maintains **state** across a task much like a model-based agent (conversation history, scratchpad memory), and can be paired with **learning** techniques (fine-tuning on its own successful trajectories).

What is genuinely new is not the taxonomy – it is that a single, general-purpose LLM can now play the role that used to require a hand-coded rule set or a narrowly trained policy for every new task. The same underlying model can be a goal-based trip planner today and a utility-based coding assistant tomorrow, with no retraining required in between.

2.9 Summary

We formalized the agent-environment loop and classified agents into five classical categories: simple reflex, model-based reflex, goal-based, utility-based, and learning agents, each adding a capability – state, planning, optimization, or learning – that the previous type lacked. We connected the learning agent category directly to the reinforcement learning concepts from last semester. Finally, we positioned agentic AI not as a new category, but as a new, general-purpose way of implementing goal-based, utility-based, and learning behavior using an LLM as the reasoning core.

Exercises

1. **Classify it.** For each system below, identify which of the five classical agent types it best matches, and justify your answer: (a) an elevator control system, (b) a chess engine that searches several moves ahead, (c) a spam filter that adapts based on emails you mark as spam, (d) a rule-based irrigation sprinkler that turns on at a fixed time each day.
2. **From reflex to learning.** Take the thermostat example from Section 2.2. Describe, step by step, what you would need to add to turn it into (a) a model-based reflex agent, and then (b) a learning agent.
3. **Utility functions.** Propose a utility function (in words, not necessarily math) for a food-delivery app deciding which rider to assign to a new order. What competing factors should it balance?
4. **Connecting to RL.** In reinforcement learning terms, what plays the role of the “reward” for a learning agent? Using the AlphaGo example, describe what its reward signal might have been.
5. **Agentic AI positioning.** In your own words, explain why agentic AI is described in this lecture as “a new implementation” of the classical taxonomy rather than a sixth category. Do you agree or disagree? Justify your answer.